CASE 9
DIGEORGE SYNDROME

The embryonic development of the thymus.

The thymus is the central lymphoid organ in which T cells develop and mature (see Cases 5 and 6). It is composed of an epithelial stroma that becomes populated with precursor T cells and other cells of hematopoietic origin such as macrophages and dendritic cells. The thymic stromal cells provide a microenvironment that is essential for the attraction, survival, expansion, and differentiation of the T-cell precursors. The thymus initially develops in the embryo as an epithelial anlage that gives rise to the thymic stroma. The thymic epithelium derives from the endoderm of the third pharyngeal pouch (Fig. 9.1). The pharyngeal arches and pouches are the embryologic segmental structures that develop into organs of the face and upper thorax. Part of the third and the fourth pharyngeal pouches give rise to the parathyroid glands, to which early thymic development is closely linked. Mesenchymal cells from the pharyngeal arches are also essential for thymic development, giving rise to the connective tissue of the thymus as well as to the smooth muscle of the heart and the major arteries.

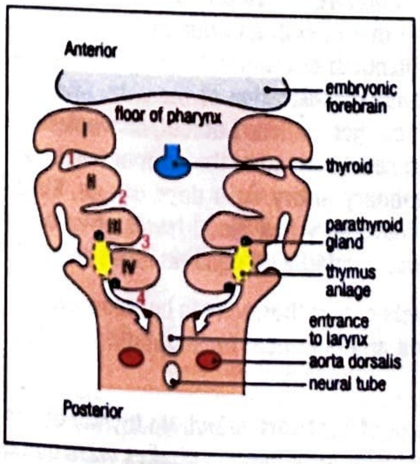

Fig. 9.1 The thymus originates from the pharyngeal pouches. This cutaway view of the embryonic pharynx shows the segmented structure of the region comprising the four pharyngeal arches (I–IV, shaded pink) on either side of the ventral midline, separated internally by pharyngeal pouches (1–4). A thymus anlage (yellow) arises from the third pharyngeal pouch on each side, in close proximity to the parathyroid glands. As the embryo develops, the two anlagen come together to form a single organ. The correct segmentation of the embryonic pharynx, which is under the control of the transcription factor TBX1, is required for the correct development of the thymus and other organs and structures that develop from this region.

TOPICS BEARING ON THIS CASE:

- Thymic development
- Thymic epithelial cells
- Immunodeficiency
- Thymus transplantation

This case was prepared by Luigi Notarangelo, MD, in collaboration with Ari Fried, MD.

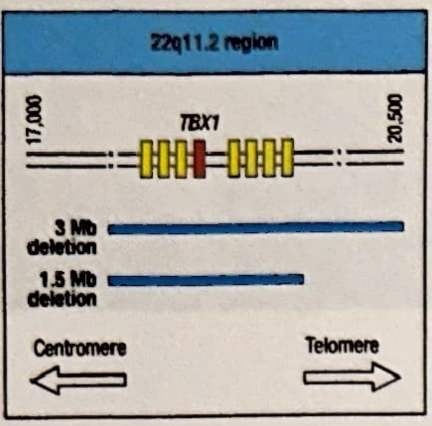

Fig. 9.2 Map of the chromosome region that harbors the 22q11.2 deletion commonly observed in patients with DiGeorge syndrome. The most common abnormality is a deletion of 3 Mb, which is found in 90% of patients with 22q11.2 deletion. A 1.5 Mb deletion is found in 8% of patients, and the remainder have smaller deletions or point mutations. Disruption of the gene encoding the transcription factor TBX1 has been identified as the major cause of the heart abnormalities, as well as of defects in the thymus and parathyroid glands that are associated with DiGeorge syndrome.

The transcription factor TBX1 has a central role in the development of the pharyngeal apparatus and its derivates, including the thymus, the parathyroid glands, and some tissues of the developing heart. TBX1 belongs to a family of transcription factors that have a common DNA-binding sequence, designated the T-box. These T-box factors have a role in early embryonic cell fate decisions and in the regulation of the development of many embryonic and extraembryonic structures. TBX1 is expressed in the endoderm of the third pharyngeal pouch and in the adjacent mesenchyme. It regulates the expression of several growth factors and transcription factors important for development of the thymus and the parathyroid glands. As a regulator of embryonic patterning, TBX1 not only controls the segmentation of the embryonic pharynx but is also required for the growth, proper alignment, and septation of the cardiac outflow tract.

As one might expect, therefore, deletion or mutation of the *TBX1* gene leads to a wide range of congenital defects, including defects in thymus development. However, *TBX1* is only one of the more than 35 genes located at chromosome 22q11.2, and *TBX1* gene defects account for a very small proportion of thymus development defects in humans. In contrast, deletion of this region in one of the two chromosomes 22 is the most common cytogenetic abnormality associated with DiGeorge syndrome (Fig. 9.2), a condition characterized by congenital heart defects, hypoparathyroidism and hypocalcemia, and a variable degree of immunodeficiency.

The case of Elizabeth Bennet: severe immunodeficiency as a result of disrupted development of the thymus.

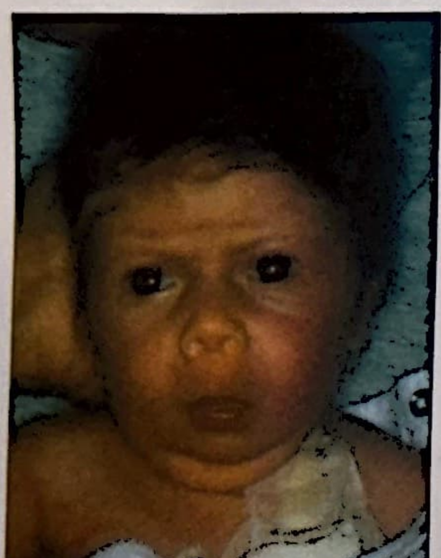

Fig. 9.3 Typical features of DiGeorge syndrome. These include low-set ears, hypertelorism (an increased distance between the eyes), and a small mouth and an underdeveloped jaw (micrognathia).

Elizabeth Bennet was born at term after an uncomplicated pregnancy. She had a low birth weight of 2.1 kg, and dysmorphic facial features were noted at birth, including low-set ears as well as a relatively small mouth with an undersized lower jaw (micrognathia) (Fig. 9.3). At 2 days of life, Elizabeth developed feeding difficulties, rapid breathing, increased fatigue, and a bluish discoloration of the skin. She was diagnosed with truncus arteriosus, a severe congenital heart defect, characterized by a single common outflow tract leaving the heart instead of the normal two separate blood vessels—the aorta and the pulmonary artery. At 4 days of age, Elizabeth developed seizures. She was found to have very low blood levels of calcium (6.2 mg dl^{-1}, normal 8.5–10.2 mg dl^{-1}) and was treated with calcium and vitamin D.

Her hypocalcemia resulted from very low levels of parathormone in her blood, a hormone that is made by the parathyroid glands and is critical for regulating calcium and phosphorus homeostasis in the body.

Elizabeth underwent cardiac surgery for repair of her heart defect. No thymic tissue was identified intraoperatively. After successful surgery, genetic studies were done; these revealed a normal karyotype, which excluded major chromosomal rearrangements. However, with the help of fluorescence *in situ* hybridization (FISH), she was

found to have a deletion of the chromosomal region 22q11.2 on one of her two copies of chromosome 22, consistent with a diagnosis of DiGeorge syndrome (Fig. 9.4).

At an immune evaluation at 2 weeks old, Elizabeth's absolute lymphocyte count was low for her age, with 560 cells μl^{-1} (normal >2500 lymphocytes μl^{-1}). She had almost no $CD3^+$ T cells, with a count of 11 cells μl^{-1}, which was less than 1% of her total lymphocyte count, while $CD19^+$ B-cell numbers and $CD16^+/CD56^+$ NK cells were in the normal range for her age. Her peripheral blood mononuclear cells (PBMCs) responded poorly to the mitogens phytohemagglutinin (PHA) and concanavalin A (ConA), which is indicative of poor T-cell function (see Case 5 and Fig. 5.7). Her significant T-cell defect led to the diagnosis of complete DiGeorge syndrome, a rare variant of DiGeorge syndrome (less than 1% of all cases of DiGeorge syndrome) that is associated with severe immunodeficiency and death if not treated early in life.

Newborn with heart defect, seizures, and hypocalcemia.

Severe T-cell lymphopenia.

Elizabeth was started on prophylactic antibiotics to prevent infection with the opportunistic pathogen *Pneumocystis jirovecii*. At 6 months of age she received a thymic transplant into her right leg quadriceps muscle. A biopsy of the thymic graft, performed a few months after transplantation, showed significant presence of thymocytes within the transplanted thymic tissue. One year after transplantation, Elizabeth had developed a significant number of T cells (860 $CD3^+$ cells μl^{-1}), although she did not reach normal T-cell counts. Her T cells responded normally to mitogens and antigens *in vitro*. After immunization, Elizabeth mounted protective antibody responses to tetanus toxoid, *Haemophilus influenzae* type b, and *Streptococcus pneumoniae* (pneumococcus). She required calcium supplementation for several months and needed repeat cardiac surgery at the age of 3 years, which she tolerated well.

Thymic transplantation, leading to improved T-cell count.

At 6 years old Elizabeth developed purple bruises (purpura) and pinpoint red lesions (petechiae) on her skin. She was found to have a low platelet count of 15,000 μl^{-1} (normal 150,000–300,000 μl^{-1}). This was due to destruction of her platelets by autoantibodies, resulting in insufficient blood clotting and therefore bleeding into the skin, a condition called immune thrombocytopenia. She was successfully treated with high-dose intravenous gamma globulin (1 g per kg body weight), which resulted in a rapid increase in the platelet count to 115,000 μl^{-1}.

DiGeorge syndrome.

A microdeletion of the 22q11.2 region on one chromosome 22 is the most commonly diagnosed cytogenetic deletion in humans, with an estimated prevalence at birth of 1:4000. It results in a variable clinical phenotype, which includes DiGeorge syndrome, velocardiofacial syndrome (VCFS), or conotruncal anomaly face syndrome. Most 22q11.2 deletions are spontaneous and arise as a result of an aberrant meiotic exchange event. Patients with both DiGeorge syndrome and VCFS have numerous, overlapping clinical features, including the absence (aplasia) or underdevelopment (hypoplasia) of the thymus, hypoparathyroidism, cardiovascular defects, and structural defects of the face and pharynx (see Fig. 9.3). Ninety percent of patients with a 22q11.2 deletion share the same 1.5 Mb or 3 Mb monoallelic microdeletion (see Fig. 9.2); a few patients have smaller deletions that result in the same phenotype.

Patients with the hemizygous 22q11.2 deletion can be easily diagnosed by FISH (see Fig. 9.4). However, nearly half of patients with DiGeorge syndrome do not carry 22q11.2 deletions. Point mutations of the *TBX1* gene have been identified in a few of these patients, and deletions on chromosome 10 in others. A phenotype similar to that of DiGeorge syndrome can also be observed in patients with CHARGE syndrome, which is due to mutations in the gene for chromodomain helicase DNA-binding protein 7 (*CHD7*). CHARGE stands for coloboma (small structural defects)

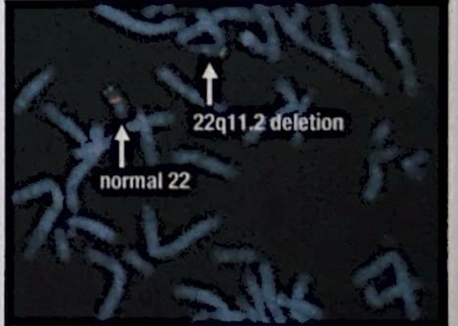

Fig. 9.4 Hemizygous deletion of 22q11.2 detected by fluorescence *in situ* hybridization (FISH). Chromosomes were stained with a green probe that recognizes the subtelomeric region of chromosome 22, and a red probe that stains the 22q11.2 region. The absence of this region is apparent in one of the two chromosomes 22. Micrograph courtesy of Sergio Barlati.

of the eye, *h*eart defects, *a*tresia of the choanae (a blockage of the nasal passages), *r*etardation of growth and/or development, *g*enital and/or urinary abnormalities, and *e*ar abnormalities. Patients with CHARGE syndrome may also present with immunodeficiency. Because of the similarities of clinical phenotype, CHARGE syndrome must be considered in the differential diagnosis with DiGeorge syndrome.

DiGeorge syndrome can encompass a broad range of clinical features, but most infants present with a congenital cardiac defect, mild to moderate immunodeficiency, facial dysmorphisms, developmental delay, palatal dysfunction, feeding difficulties, and hypocalcemia due to absent or low function of the parathyroid glands. Neurobehavioral and psychiatric abnormalities (schizophrenia) may be observed in a significant fraction of patients, especially during adolescence or adulthood.

The degree of immunodeficiency in DiGeorge syndrome is highly variable and does not correlate with the presence or severity of other clinical features of the syndrome. Most patients have a small thymus and a milder immune defect, characterized by a mild to moderate decrease in T-cell counts, but intact T-cell function. These patients are classified as having 'incomplete DiGeorge syndrome.'

At the severe end of the spectrum are patients with complete absence of a functional thymus and profound T-cell lymphopenia. These patients represent less than 1% of all patients with DiGeorge anomaly, and are given the diagnosis of 'complete DiGeorge syndrome.' They have severe combined immunodeficiency with increased susceptibility to opportunistic infections and tend to die by the age of 1–2 years if not treated adequately. To restore T-cell function, infants with complete DiGeorge syndrome have to undergo thymic transplantation. Patients with complete DiGeorge syndrome and a minority of those with incomplete form of the disease can be recognized at birth because they have a low number of T cell receptor excision circles (TRECs), which are enumerated during newborn screening for SCID (see Fig. 5.4).

In some cases, patients with complete DiGeorge syndrome may develop expansion of a small number of clones of T cells and display severe skin rash and lymphadenopathy, leading to a phenotype that resembles Omenn syndrome (see Case 7). In this subgroup of patients (known as 'atypical complete DiGeorge syndrome'), variable (or even increased) numbers of circulating T cells (expressing the activation marker CD45R0) are detected, and T lymphocytes infiltrate the skin and other organs. Autoimmunity, especially leading to a reduction in blood cells (cytopenia), is another sign of immune dysregulation that is frequently observed in patients with DiGeorge syndrome, even after successful thymic transplant, as seen in Elizabeth's platelet deficiency.

Mutations of the *FOXN1* gene, which encodes a transcription factor essential for thymic epithelial cell development, account for another rare immunodeficiency with very severe T-cell lymphopenia. These patients also present with generalized alopecia and lack of hair follicles. This condition represents the human equivalent of the 'nude' mouse phenotype.

Questions.

1 *Bone marrow transplantation is the treatment of choice in other cases of severe immunodeficiency. Would it work in patients with DiGeorge syndrome and in patients with* FOXN1 *deficiency?*

2 *Thymic transplantation in patients with DiGeorge syndrome is performed using thymic tissue from an unrelated donor, who typically is not HLA-matched to the patient. How can the transplant be successful?*

3 *How could you explain why most patients with DiGeorge syndrome do not have a very severe T-cell deficiency, and that they tend to normalize calcium levels with time?*

4 *What could be the reason that patients with DiGeorge syndrome are at higher risk of autoimmune manifestations?*